

Related Work Map for Angular Manifold Routing

Preprint

This document summarizes the most relevant research areas adjacent to the Angular Manifold Routing architecture. The goal is to position the work clearly in a preprint while acknowledging prior literature. The central observation motivating this research is that routing tokens by angular sector on a normalized hypersphere appears to produce sublinear routing occupancy. Empirically this has the approximate form: $E(K) \approx c * K^\alpha$, where $\alpha < 1$. This produces automatic sparsity and potential hardware locality improvements without requiring a learned gating network. The sections below summarize the closest conceptual neighbors in current research.

Hyperbolic and Non-Euclidean Representation Learning

Research in hyperbolic and non-Euclidean neural networks explores representing hierarchical data structures using curved spaces. Hyperbolic embeddings are particularly useful for representing taxonomies, graphs, and hierarchical language structure. Closest overlap: the recognition that representation geometry influences efficiency. Key difference: these works study representation structure rather than routing architectures. They do not introduce deterministic angular routing partitions or measure routing capacity scaling.

Standard Mixture-of-Experts Routing

Modern mixture-of-experts systems route tokens using learned gating networks. A routing function scores experts and selects a small subset for each token. Closest overlap: conditional computation and sparse routing. Key difference: routing is learned during training rather than emerging from geometric partitioning. Load-balancing losses are typically used to enforce uniform expert utilization.

Hyperspherical MoE Research

Some work explores routing computations on normalized hyperspherical embeddings. Token and expert representations are normalized and routing scores are computed using cosine similarity. Closest overlap: hyperspherical geometry inside routing computations. Key difference: routing still depends on learned gating scores. Geometry is used as a scoring space, not as a deterministic sector-partition routing mechanism.

Spherical Routing Variants

Recent research has proposed spherical routing ideas in domain-specific systems such as image restoration. These systems activate experts based on similarity relationships in spherical embedding spaces. Closest overlap: explicit spherical routing language. Key difference: expert activation is still controlled by learned similarity scores rather than fixed angular sector partitions. These works also do not analyze routing occupancy scaling behavior.

Geometry-Aware Aggregation

Some research focuses on preserving geometric relationships when combining outputs from multiple experts. Vectors may be decomposed into angular and radial components on a sphere or other manifold. Closest overlap: explicit angular and radial geometry modeling. Key difference: these techniques modify aggregation behavior rather than the routing mechanism that determines which experts are activated.

Latent Space Routing Improvements

Other recent work attempts to improve routing stability by learning better latent routing spaces. These systems often impose low rank structure or Lipschitz constraints on routing functions. Closest overlap: the idea that routing depends on geometric structure of embeddings. Key difference: routing remains a learned transformation rather than a deterministic geometric partition.

Area of Prior Work	Main Idea	How It Differs From Angular Manifold Routing
Hyperbolic Representation Learning	Curved spaces represent hierarchical relationships efficiently	Focuses on representation geometry rather than routing architecture
Standard MoE Routing	Learned gating networks select experts	Routing is learned rather than determined by geometric partition
Hyperspherical MoE Methods	Normalized embeddings with cosine similarity routing scores	Geometry used for scoring rather than deterministic sector routing
Spherical Routing Variants	Domain specific spherical expert activation methods	Still relies on learned similarity rather than fixed sectors
Geometry-Aware Aggregation	Preserves angular and radial structure during expert aggregation	Affects output aggregation rather than routing decisions
Latent Routing Space Methods	Learns improved routing geometry in latent space	Routing still determined by learned functions

Suggested positioning statement for a preprint: While prior work explores geometric representation learning and mixture-of-experts routing, we are not aware of work that combines normalized hyperspherical embeddings, deterministic angular partition routing, emergent routing sparsity through bucket compression, and an empirical routing occupancy scaling law with hardware locality implications.